

Abhishek Tjo Andrew Prasanth

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Professional Summary

Mechanical Engineering student (4.0 GPA) with hands-on experience in CAD modeling, dynamic controls testing, and embedded mechatronic systems. Proficient in Autodesk Inventor, rapid prototyping, and sensor integration using Python, Arduino, and oscilloscopes. Seeking a Summer Robotics or Mechanical Engineering Internship to apply hardware prototyping and control systems skills.

Education

Anderson University

August 2024 - May 2028

Bachelor of Science (BS), Mechanical Engineering

GPA: 4.0

- **Honors & Awards:** Dean's List, Honors Program Scholar
- **Relevant Coursework:** Kinematics & Robotics, Signals & Controls, Dynamics, Solid Mechanics, Digital Electronics, Circuit Analysis, Differential Equations, Linear Algebra

Professional Experience

Flight Controls & Hardware Co-op | Leo Flight / Anderson University

May 2024 - Present

Intern

Indiana, USA

- Designed and 3D printed a physical gyroscope test rig, incorporating precise dimensional tolerances for flight stabilization testing.
- Built a real-time motion telemetry tool using Python to track dynamic controls and flight stability metrics.
- Analyzed SBUS protocol controller signals and debugged Arduino hardware interfaces using a digital oscilloscope.
- Hand-soldered wiring harnesses and integrated Electronic Speed Controllers (ESCs) for power delivery and motor regulation.

Projects & Outside Experience

Engineering Mission trip

January 2026 - May 2026

Costa Rica

- Calculated electrical power load profiles and evaluated solar irradiance data to design an off-grid PV system optimized for tropical field conditions.
- Simulated circuit behavior using LTspice and drafted standardized system schematics to enable modular replication across future community installations.
- Managed project budget and field logistics to source components, build hardware, and present technical proposals to local leadership.

Prosthetic / Robotic Hand Mechanical Design

July 2026 - Present

- Engineered 3D CAD models and mechanical assemblies in Autodesk Inventor, refining finger joint kinematics and linkage movement for realistic articulation.
- Designed schematic and circuit layout for robotic arm power distribution.
- Developed motion tracking control code in Python to process sensor inputs and command multi-joint articulation.

Putt-Putt Course

- Designed and fabricated an interactive putt-putt course using AutoCAD, rapid prototyping (3D printing), and shop machinery, securing 2nd place out of the entire course cohort.
- Integrated Arduino microcontrollers and electromechanical actuators to power custom dynamic obstacle features, including automated opening doors and timing mechanisms.
- Collaborated with a teammate to execute full project lifecycle, from initial CAD modeling and tolerance fitting to final hardware assembly and embedded control testing.

Skills

- **CAD & Mechanical Design:** Autodesk Inventor, AutoCAD, Kinematics, Linkage Design, Technical Drawing Generation
- **Embedded Systems & Hardware:** Arduino, SBUS Protocol, ESC Integration, KiCAD, Digital Oscilloscopes
- **Software & Controls:** Python, Java, LTspice, C, C++
- **Prototyping & Fabrication:** Machining, Hand Soldering, Rapid Prototyping (3D Printing)
- **Languages:** German, Hindi, Tamil, Malayalam, English

Certifications

Python (Scope India), Arduino & Robotics (MyRobo Trivandrum, India)